NeuRoad's Problem

How do you usually start your day? Does it start with waking up, brushing your teeth, eating breakfast, getting to work and enjoying the rest of your day, typically, with minimal stress and to fully focus on achieving your goals? Or whenever you have a meeting with someone, do you try to rush there as early as possible? You smile and greet everyone just like how everyone else does. However, for some people, their methods of completing daily tasks and social interactions may be viewed as different or out of the ordinary. This is generally a sign of being Neurodivergent, however, this might not be applicable to everyone, as neurodiversity is a spectrum and there are many types of it including dyslexia, ADHD, autism and many more. People may often have a stigmatised or negative perception of neurodiversity, as a result of their lack of awareness on the subject.

Research states that 1 in 5 individuals are a part of this community, so being neurodivergent is more common than you think *(Doyle, 2020)*. Most of them are not as far off on the spectrum as society may believe. Some of your friends may also be neurodivergent, demonstrating that people under the spectrum can act exactly or similarly to you! It's just that their way of processing and coping with the world may be misunderstood. They might struggle with some tasks but when being put in the right environment and receiving understanding from others, they will be able to shine the brightest in their specialty.

Once a person understands that they may be neurodivergent, usually, the next step is to see a specialist and grow an understanding of what they're dealing with, as well as, receiving help and guidance. However, in many countries, communities do not have access to the appropriate health service and receive this type of support. In other cases, individuals themselves do not have the financial means to afford going to a specialist. These factors hinder neurodivergent individuals' ability to truly shine, especially in areas where neurodiversity is seen as a stigma.

The solution

This is where our new solution comes into play. NeuRoad, is an AI incorporated application that strives and aims to help offer neurodivergent communities more affordable and accessible alternatives, instead of going to a medical professional that may not be a viable pathway for most individuals. Especially, as our team is located in Vietnam. Neurodivergent children, especially those who are in the rural areas of Vietnam are not accepted by their families. Cultural stigma and limited awareness can influence Vietnamese families to seek out for experts' support.

Alone. Abandoned. Neglected. Children with neurodiversity struggle even more in those settings. In the worst case scenario, they would be sent off to asylums as they are seen as being "crazy" or "sick". These children would not get the chance to even

be loved, understood and seen. They would not have access to education and a safe environment where they can flourish and develop. Children are seen as the seeds of the future but without the love and care from their parents, their abilities and potential will be constrained and limited.

By introducing platforms such as NeuRoad to these families, this helps educate them about the struggles that their children are going through with the aim of destigmatising individuals with learning needs. Additionally, it provides pathways for these families to help their child ; they may not afford going to a specialist, however, using this app for free to get additional support is a more feasible idea and only a small extra fee for additional features of the app which can aid in the experience when using the app.

NeuRoad isn't just for parents with children with learning needs, it is also for the individual themselves who are struggling, and even teachers / developmental specialists who may need an extra platform to offer support. NeuRoad is for all and we strive to develop an app that is inclusive and supportive for all individuals.

NeuRoad's Unique Selling Point (USP)

Currently, there are numerous apps and websites also targeting the challenges that neurodivergent individuals might face. For instance, BetterHelp is an app that helps these individuals match with a therapist who suits their needs. Users have the option to communicate with their therapist via text, live chat, audio and video sessions.

Other companies are also utilising the same method as a solution for this problem. Nevertheless, these sessions need to be subscribed monthly which might not be an option for many families in Vietnam and in developing countries, in general. Furthermore, these applications focus more on helping individuals find a therapist, rather than giving them direct and real life solutions. When working with therapists, individuals need the ability to express themselves verbally and clearly, for the therapists to understand their struggles and come up with solutions. This can disregard individuals on the spectrum who are unable to express their opinions verbally. Consequently, excluding them and ignoring the ultimate goal of these apps which is to be inclusive and acknowledging diversity.

How NeuRoad works

In the beginning as you open the app you'd first see a survey/form that you'd have to fill out. It would be a series of questions starting from who is the user, the individual itself or a teacher for example, followed by a pop up to state if you are diagnosed or not. If you are not diagnosed, a series of questions may appear to determine what the individual may struggle with and their strengths. This will be stored in the AI's data base for future reference. Proper diagnosis will not be done as this is not run by a proper specialist rather it is supported by them.

After the form has finished, the users will then have full access to the other features of the application, including an AI chat box with preset prompts that may be useful, news content that may be educational for users, and a daily genie that is customized to each user's needs. Examples of this function are, for users who struggle with organisation, there can be a task daily reminder, that can also break down larger chores into a checklist of smaller and easier tasks. Users that may struggle with dyslexia can be reminded by a genie for a daily practice typing to improve their spelling or parents that want to help their neurodivergent child become as self sufficient as possible by learning daily skills/chores, and so on.

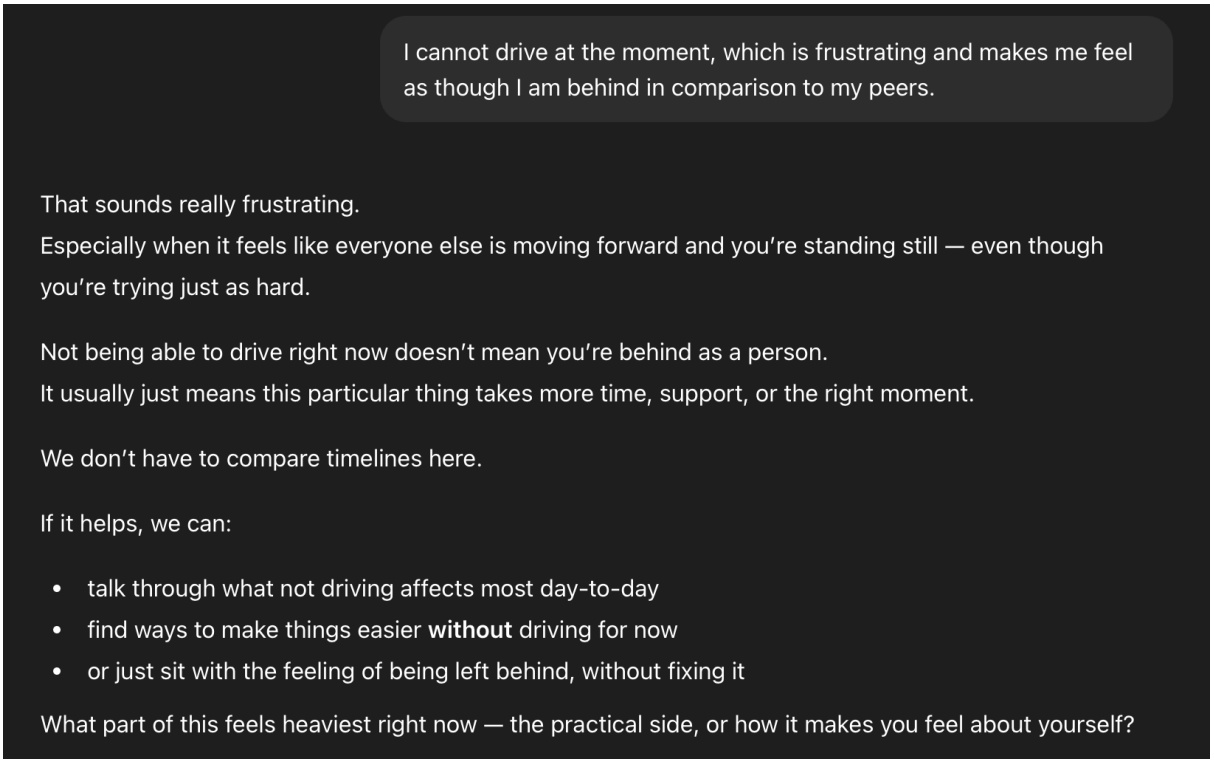
As a way to encourage individuals on the spectrum who struggle with expressing their ideas, there would be an option of selecting the tone/style that the chatbox delivers their messages in. This allows the chatbox to adapt to any user, helping them feel more comfortable sharing their thoughts on the app. For example, the chatbox can give suggestive feedback and add on more recommendations to an idea that has already been sent by the user.

How NeuRoad ensures users' safety:
We will make sure users feel comfortable and secure when using NeuRoad. The users will not be asked to use their real names but nicknames and all of their data will be kept confidential to the individual. In addition, before using the app, users will have to complete a Parental Confirmation form. In alarming situations when users mention self-harm, the system will be sending an alert to the parents/guardians to make sure that the users are stopped from harming themselves as soon as possible.

Evidence
Neurodivergent individuals are already finding support by using AI such as the app Neurotranslator, which helps them to communicate and understand others' behaviours . With Neuroad, individuals now do not need to enter complex prompts as the AI in our app is embedded with specialist's knowledge, which allows the experience to be more customised and emotionally safe.

To test the feasibility of our idea, we have fed an existing AI app with information about neurodiversity and stories of neurodivergent individuals. We have added stories of different neurodivergent conditions such as: autism, ADHD, tourette's, dyslexia, etc. The AI was able to develop a persona which is appropriate for these types of users, generated prompts and rules which it has to follow to make this a safe space for the users. To test the response of AI, we have used a true story from online sources and entered in what they find difficult.

This is an example of how the AI would respond to a user's prompts after reading and studying the behaviour of people with dyspraxia.



In this example, the tool names the emotion of the user without intensifying or questioning it, which shows that it is able to show sympathy towards the user, which is crucial as it creates a friendly and non-judgemental tone. Furthermore, the AI does not assume intelligence and effort of the user. It avoids diagnosing the user and gives users time and different options to choose from, instead of instructing and commanding the user. By maintaining a calm, steady tone and the use of short sentences, it can help the user feel more safe and comfortable as the user does not have to deal with complex and overloading information.

Further development

In the future, when our app has successfully been implemented and receives good feedback from users, we would add on additional features which can help our app engage more with the users.

There can be an optional feature of gaining points by completing set targets tailored to each individual. These targets are tasks that the chatbox or the individual has suggested in order to help them complete daily tasks more efficiently or help them improve difficulties they might face because of their neurodivergence. The points will be added and they will be ranked on a leaderboard. This can not only make it more fun and engaging, it can also help individuals share their stories with others with similar neurodivergence, allowing them to feel more included and welcoming in a community.

In addition to the scoring points function, the user can also have an option to choose a virtual pet/animal that they can take care of. The users can trade their points for their pet essentials such as food, shelter, clothing, etc. Research has demonstrated the effectiveness of virtual pet therapy at helping autistic children learn *(Altschuler, 2007)*. Having a pet to take care of allows individuals to imagine the pet's needs and wants, "putting themselves in others' shoes". Through these interactions with their pets, neurodivergent individuals are encouraged to think about others and engage in social interactions.

How NeuRoad will be sustained

We will be asking for funds from multiple companies, local organisations, education centres and advertise small icons/pictures of their organizations on the app. The app will have a dedicated space for sponsors, making the experience of using NeuRoad more enjoyable. Additionally, the app will also have an option to receive donations, a part of this money will go to sustaining our app and a portion of it will go towards a mental health organisation.

Financial breakdown

In the beginning of the developmental stage of this app the core development and app design costs would be free as our team is equipped with the skills to code the basic functions of this app. The main driving cost in the beginning would be the cost to integrate AI which is around $20,000 - $50,000.

Below is a table that breaks down the costs of features and maintenance that would be added in the future when this app goes beyond the development stage. These costs were estimated through comparison to an app with similar functionality such as Duolingo.

Type of cost	Cost in dollars
Addition of Game-like features	~ $10,000 – $30,000
Integration of speech recognition	~ $10,000 – $25,000
Quality assurance and tests	~ $3,000 – $5,000
Total costs of adding advanced features	~$23,000 – $70,000
Maintenance and updates	~ $2,000 – $5,000+ / month
Storage of user cloud	$0.023 / GB / month (frequent access provider)
Cloud protection	~ $500 – $5,000 / month
Total monthly costs	~$3,000 – $15,000